BEFORE THE LESSON · HOME TASK · 10 MINUTES

Geometric Sequences and Series

Multiply by the same number each step, then add the results with one formula

Why you are doing this: we start the lesson with an 8-question Wayground check. This page tells you exactly what it will ask, so you can walk in ready instead of guessing.

Mr Malek Thiab | Grade 11 | Lesson 6-7 | Week 8 · Semester 1, 2026–27

WATCH FIRST · VIDEO

Multiply by the Same Number, Every Step

A geometric sequence has a common ratio; a geometric series adds its terms

YOUR VIDEO · YOUTUBE

What Are Geometric Sequences and Series?

Channel: The Organic Chemistry Tutor

https://www.youtube.com/watch?v=zRKZ0-kOUZM

Watch the parts on the common ratio, the nth term and the sum of a finite series.

THE ONE IDEA

In a geometric sequence every term is the one before it times the same common ratio. That gives a direct formula for any term, and when the ratio is not 1 the sum of the first terms has a closed formula too.

$$a_n = a_1\, r^{\,n-1}$$

$$S_n = \frac{a_1\,(1 - r^{\,n})}{1 - r} \quad (r \neq 1)$$

Exponent rules

$$5 \cdot 3^4 = 5 \cdot 81 = 405$$

Do the power first, then multiply.

Common ratio

$$r = \frac{a_2}{a_1} = \frac{12}{6} = 2$$

Divide a term by the term before it.

Sigma notation

$$\sum_{k=1}^{4} k = 1 + 2 + 3 + 4 = 10$$

Add the summand for each index value, from the starting index to the ending index.

STUDY THIS · ONE EXAMPLE

One Example, Every Step Shown

For 6, 12, 24, and so on: find the 6th term and the sum of the first 6 terms. Cover the right-hand column, try it, then check.

1	$6,\ 12,\ 24,\ \ldots \quad a_1 = 6, \quad r = \dfrac{12}{6} = 2$	Name the first term, then divide a term by the one before it to find the common ratio.
2	$a_n = 6 \cdot 2^{n-1}$	Write the explicit formula: the power is the term number minus 1, because the first term has not been multiplied yet.
3	$a_6 = 6 \cdot 2^5 = 6 \cdot 32 = 192$	Substitute the term number 6, so the power is 5. Check: 6, 12, 24, 48, 96, 192.
4	$S_6 = \dfrac{6\,(1 - 2^6)}{1 - 2} = \dfrac{6(-63)}{-1} = 378$	Sum of the first 6 terms with the sum formula. Here the power is the number of terms, 6. Check by adding: 378.

The mistake most people make: using the term number as the power. For the 6th term that gives 384, which is really the 7th term: the power is one less than the term number.

READINESS CHECK · 3 MINUTES

Four Questions: Answer Before Class

This is not graded. It tells me what to teach first. An honest answer helps you more than a lucky one.

1 $80,\ 40,\ 20,\ \ldots$: find r

2 Evaluate $2 \cdot 3^4$

3 Write out and add: $\sum_{k=1}^{4} 2k$

4 In one sentence: how can you tell a geometric sequence from an arithmetic one?

HOW TO SUBMIT

- Type your four answers into the LMS form, one line each.
- If you are stuck on one, write “not sure yet” and say what confused you. That is a useful answer.
- Due: before the bell. It takes three minutes.

In class we start with: an 8-question Wayground kick-off on exactly these ideas: exponent rules, the common ratio from terms and from a graph, sigma notation and the sum formula. Then we write formulas for geometric sequences.